Abdulaziz Yusupaliev

Machine Learning Engineer · speech recognition, LLM applications, Python backends

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Machine learning engineer at Uzinfocom AI Labs. I trained and published a Karakalpak speech-to-text model (18.3–18.6% WER) and build what puts models to work: retrieval and agent pipelines, typed APIs, and evaluation.

EXPERIENCE

Machine Learning Engineer · Uzinfocom AI Labs (zehnmind.ai) — Sep 2026 – Present

- Review and correct transcriptions that become speech-to-text training data.
- Annotate and quality-check speech datasets for ASR models.
- Joined the core team after a three-week internship.

AI/ML Intern · IT Park — Apr 2026 – Aug 2026

- Cleaned, encoded, and scaled datasets with pandas and scikit-learn transformers.
- Built and evaluated scikit-learn pipelines for regression and classification (MSE, MAE, R^2, accuracy).

Frontend Engineer · Hiwelcome — Jan 2026 – Present

- Build pages and reusable components on the company's APIs with Next.js, TypeScript, and Tailwind CSS.

PROJECTS

Karakalpak Speech Recognition · published on Hugging Face — 2026

Python · PyTorch · Transformers · wav2vec2 XLS-R · LoRA · KenLM · pyctcdecode · librosa · jiwer

- Fine-tuned wav2vec2 XLS-R 300M with LoRA on 77.45 h (17,685 clips) of a low-resource language, about 2.5 hours on one GPU.
- 4-gram KenLM decoding: 18.61% WER on news, 18.30% on literary speech (from 38.37%, 22.32% greedy); 20.3–21.3% external.
- Validated 22,108 recordings with 37 automated checks; wrote Karakalpak text normalization and a 35-token vocabulary.
- Released the model, decoder, and datasets with a CPU inference client.

Multi-Agent AI Analyst · github.com/vvadr/Multi-Agent-AI-Analyst — 2026

FastAPI · LangGraph · Gemini · Qdrant · PostgreSQL · SQLAlchemy · Redis · Docker · GitHub Actions

- LangGraph supervisor routing retrieval, web, read-only SQL, and critic agents to a cited answer.
- Durable runs queued in PostgreSQL and streamed over SSE, resumable after a restart or from another device.
- Tenant isolation, Argon2 auth, rate limits; 27 pytest modules, CI, and RAGAS/Langfuse evaluation hooks.

LoanSource AI · LLM fine-tuning — 2026

Qwen2.5-1.5B · QLoRA · PEFT · TRL · bitsandbytes · Gradio

- Fine-tuned Qwen2.5-1.5B with QLoRA on a free Colab T4 for loan-document Q&A, with retrieval supplying the citations.

Loan Terms Assistant · RAG — 2026

FastAPI · Gemini · Qdrant · Next.js · Docker

- Answers from one loan document with a page citation, checked against the retrieved clauses before it is shown.

Yelp Restaurant Rating Pipeline · data pipeline — 2026

pandas · scikit-learn · Parquet · joblib

- One-command pipeline from raw Yelp data to a 22-feature dataset of 5,852 restaurants; sentiment led feature importance (0.79).

PRACTICE PROJECTS

Customer Churn Preprocessing · pandas, scikit-learn — missing values, TotalCharges cleanup, encoding, leakage-safe pipelines

K-Means Clustering · scikit-learn, Matplotlib — grouped similar data points with K-Means and visualized the clusters

SKILLS

Machine learning	Python, pandas, NumPy, scikit-learn, Matplotlib, Seaborn, Jupyter
Deep learning & speech	PyTorch, Transformers, PEFT (LoRA, QLoRA), TRL, wav2vec2 + CTC, KenLM, librosa, Hugging Face Hub
LLM applications	RAG, embeddings, LangGraph, Gemini API, Qdrant, RAGAS, Langfuse
Backend	FastAPI, SQLAlchemy, Alembic, PostgreSQL, Redis, NestJS, Docker
Delivery	Git, GitHub Actions, pytest, ruff, Render, Vercel; TypeScript and Next.js for product UIs

EDUCATION

Full Scholarship Program in AI & ML · Bepro, World Bank & IT Park — pipelines, regression, trees, SVM/KNN, model selection — 2026

Frontend Development Program · IT Park — React, Next.js, TypeScript — 2024

Frontend Development Program · Proweb — HTML, CSS, JavaScript — 2022 – 2023

Recognition: National and international AIMO medalist.